

3.5

2. Three independent vectors

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$-4 \quad \begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Basis = $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$n \in \mathbb{R}^1$, not $\mathbb{R}^3$

Intersection with $z=0$:

Line in plane $\Rightarrow \begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

Basis = $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Perpendicular:

$$\begin{bmatrix} -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$\Rightarrow$ Basis: $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

37. $A = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & c \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = AS$$

$$= \begin{bmatrix} 0 & a & b \\ 0 & 0 & a \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & c \end{bmatrix} = SA$$

$$= \begin{bmatrix} 0 & a & b \\ 0 & 0 & a \\ 0 & 0 & 0 \end{bmatrix}$$

The subset of matrices that commute with the shift S has dimension

$\boxed{3} \leftarrow 3$ independent columns

41. $M_1 \downarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, M_2 \downarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, M_3 \downarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$\times 1 \downarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times 1$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\downarrow$ Should've used P 's $\parallel$

$\Rightarrow M_3 + M_4 + M_5 - M_1 - M_2 = I$

44. The 3 dimensions must be

r, nr , and nr

$\Rightarrow r + (nr - r) = n$

3.6

11. a. No free variables

$\Rightarrow r = n, r \leq n$

n -size must be at least r

$r \leq m$

Abbreviated

(Coefficient sol. included all!)

b. $A^T y = 0$ has solutions other than $y=0$ since

$r = m$ and $r \leq n$

therefore there must be either free

variables or one solution.

* Since there is no P such that $P_i \cap P_j = \emptyset$, no permutation is dependent. Ergo, all perms have n elements in unique positions relative to other P .

24. $A^T y = d$ is solvable

when $d \in C(A^T)$; the row space of A .

The solution y is unique when the left nullspace contains 0 .

$N(A^T) = \{0\}$

-8. 20 (Not counting B/C)

B has a rank of 2:

$C(B) = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$

are the only pivot rows/columns.

Similarly, C has a rank of 2:

$C(C) = \begin{bmatrix} r & n & b & a & k & b & n & r \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

assuming all variables differ.

$N(C)$ would quite literally be equivalent to $N(C^T)$ except for the alternating 1's

$\forall i=3$

31. $x = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$AX = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

The dimension of $C(A)$ is 1.

The dimension of $C(B)$ is also 1.

(b) is also 1.

If B produces some subsequence, then it is a multiple of v and $v^T \in$ of A

$N(B^T) = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

* 4 more with alternating

-1's and remaining 0's

$\downarrow$ lined up one-by-one

$N(C^T) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

$\downarrow P=0?$ other uses 2.

a. $X = \begin{bmatrix} a & b & c \\ c & b & a \\ a & b & c \end{bmatrix}$: Capitalize

b. which matrix remains unchanged?

$X = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Retain!

By the counting theorem:

$(n-r)r = 9$ $\Rightarrow$ mem.

